

VALERIE ONSONGO

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SUMMARY

Results-driven professional with a strong passion for artificial intelligence, automation and data analytics. Adept at applying advanced machine learning techniques, data engineering, and data analysis to streamline processes and generate actionable insights. Proficient in Python, SQL, and Big Data technologies, with extensive experience in data analysis and visualization tools. Committed to delivering innovative solutions that align with organizational goals and enhance data-driven decision-making in data science, data analysis and data engineering.

EXPERIENCE

Data Scientist

SAFARICOM PLC

January 2025 - Present, Nairobi

- Developed and deployed a microservice for automated data extraction from PDFs to JSON, and conducted advanced experimentation with LlamaParse, OCR, and GENAI to handle non-selectable and low-quality documents.
- Currently leading end-to-end data analytics initiatives for a utility-sector partner, delivering actionable insights by analyzing energy consumption patterns, customer behaviors, and segmentation to enable early detection of anomalies and inefficiencies.
- Applied machine learning, anomaly detection, and time-series forecasting techniques to identify abnormal usage trends, support fraud detection, and strengthen compliance monitoring.
- Designed dashboards and rule-based alerting systems for real-time anomaly monitoring, significantly improving operational decision-making, compliance oversight, and exception handling.
- Collaborated cross-functionally to implement data-driven strategies that enhance monitoring, reporting, and risk assessment practices.

Data Scientist

Prospect 33| Global Data Lab

June 2024 - September 2024, New York

- Anomaly Detection Algorithms: Implemented advanced anomaly detection algorithms from the PyOD library on ADbench datasets, enhancing outlier detection capabilities in financial datasets.
- Data Pipeline Development: Developed a Python-based data pipeline to ensure continuous data flow, optimizing the model building and evaluation processes.
- Fraud Detection Analysis: Conducted in-depth analysis of credit card fraud detection, created robust anomaly detection models, and explored explainability techniques such as SHAP for improved model interpretability.
- Produced detailed weekly progress reports and delivered an end-of-project presentation of findings to stakeholders implementing anomaly detection in financial services.

Data Science and Automations Intern

SAFARICOM PLC

November 2023 - May 2024, Nairobi

- Continuous Monitoring Automation: Automated Oracle SQL databases for M-PESA and customer information extraction. Implemented real-time M-PESA transaction monitoring using Python-based automation, ensuring accurate tariff charging across 50+ tariffs. Enhanced financial security and customer experience with real-time notifications and exception handling, increasing stakeholder efficiency by 50%.
- Exception Reporting and Analysis System: Developed an exception reporting system using SQL and Qlik Sense. Automated report generation for improved monitoring and issue resolution. Created interactive dashboards for real-time insights and decision-making, significantly reducing manual efforts by 80% and streamlining processes for crucial EXCO presentations.
- Insights Portal and Dashboard Development: Designed and presented Qlik Dashboards for effective data translation and informed decision-making. Built a specialized audit Insights Portal, providing trending issue insights and root cause analysis for proactive issue resolution.

Data Scientist

Omdena- Nairobi Chapter

August 2023 - November 2023, Nairobi

- Unemployment Rate Prediction: Led a team in analyzing data and developing predictive models for Kenya's unemployment rate, achieving 90% accuracy using advanced algorithms like CatBoost and Random Forest. Collaborated with 10 professionals to address and provide actionable insights on Kenya's unemployment challenges.

Business Analyst Intern

CAMUSAT KENYA

September 2022 - December 2022, Nairobi

- Project Planning and Analysis: Led project planning, task assignment, and budgeting while conducting in-depth project analysis to identify improvement areas and ensure efficient resource allocation.
- Collaborated with cross-functional teams to align projects with organizational goals.
- Dashboard Development and Reporting: Developed and maintained operational and HR dashboards using Power BI, delivering real-time insights and visual reports that enhanced data-driven decision-making and improved process transparency.

PROJECT

Smart Finance Advisor

- Developed a responsive web platform using HTML, CSS, and JavaScript, enabling users to query stock prices and view interactive Chart.js-based trends. Integrated Yahoo Finance and Alpha Vantage APIs for real-time data and built an LLM-powered advisory tool using the LLAMA API for personalized financial guidance.
- Currently evolving the project into an intelligent AI agent system with LangChain and Groq's LLaMA 3, enabling tool-augmented reasoning, vector database retrieval, and structured financial decision support. The agent integrates custom financial tools for tasks like P/E ratio

calculation and historical trend analysis. Future plans include portfolio analysis with ML and real-time stock alerts.

Spotify Data Engineering Project

- Developed an end-to-end data engineering solution using AWS services, including S3, Glue, and QuickSight.
- Applied data modeling principles to design and optimize data schemas, ensuring efficient data storage and retrieval.
- Set up S3 buckets for staging and warehousing, built ETL pipelines with AWS Glue, and managed schema with Glue Crawler and Data Catalog.
- Documented and optimized data architecture through detailed diagrams, enhancing the overall data pipeline structure and performance.

Credit card fraud detection system

- Developed a credit card fraud detection system with 95% accuracy through deep data analysis and advanced algorithms like Random Forest, XGBoost, and LightGBM, leveraging rigorous K-fold cross-validation for optimal model selection.
- Deployed the system using two methods: a Flask-based web application for usability and a scalable solution using Amazon SageMaker.
- Experimented with Amazon Fraud Detector to explore enhanced fraud detection capabilities and improve system robustness.

Customer Churn Prediction Model

- Engineered a customer churn prediction model with 90% accuracy, leveraging advanced machine learning techniques and in-depth data analysis.
- Improved model performance through feature engineering and hyperparameter tuning, deriving actionable insights for customer retention strategies.
- Deployed the model using Amazon SageMaker, ensuring scalability and integration into production environments.

Loan Default Prediction Model

- Engineered a loan default prediction model using Random Forest, XGBoost, and deep neural networks, achieving 85% accuracy.
- Performed hyperparameter tuning using GridSearchCV to optimize model performance and achieve precise predictions.
- Implemented stacking techniques to combine models into a powerful hybrid system, further enhancing predictive accuracy.

EDUCATION

BACHELOR OF SCIENCE IN ELECTRICAL AND TELECOMMUNICATION ENGINEERING

MULTIMEDIA UNIVERSITY OF KENYA

APPLIED DATA SCIENCE LAB| WORLDQUANT UNIVERSITY

- Completed 8 diverse data science projects, solving real-world problems and showcasing proficiency in varied analytical techniques.

NANODEGREE- AI PROGRAMMING WITH PYTHON | UDACITY

- Completed 8 diverse data science projects, solving real-world problems and showcasing proficiency in varied analytical techniques.

SIMPLILEARN - APACHE SPARK

- Completed a comprehensive course on Apache Spark, covering fundamental concepts and practical applications, including data processing, real-time analytics, and machine learning using Spark MLlib.

HIGH SCHOOL - ALLIANCE GIRLS HIGH SCHOOL

SKILLS

Python Programming

Automation & Web Scraping: Streamlining workflows and extracting data efficiently.

Machine Learning & Deep Learning: Proficiency in TensorFlow, Keras and scikit-learn; experienced with CNNs, neural networks

Deep learning solutions: Skilled in Natural Language Processing (NLP), anomaly detection and image recognition

Data Analysis & Visualization: Skilled in Qlik Sense, Tableau, Power BI, Seaborn and Matplotlib.

SQL Database Management: Skilled in Oracle, MySQL, and PostgreSQL.

Big Data Technologies: Proficient in PySpark, Databricks, Scala and Apache Spark

Cloud Platforms: Experience with AWS services (S3, Glue, QuickSight, Athena, Sagemaker, Fraud detector)

Large Language Models & Generative AI